

Please check that this question paper contains 09 questions and 02 printed pages within first ten minutes.

MORNING

[Total No. of Questions: 09]

01 MAR 2024

[Total No. of Pages: 02]

Uni. Roll No.

Program: B.Tech. (IT)

Semester: 3rd

Name of Subject: Digital Circuits and Logic Design

Subject Code: ESIT-101

Paper ID: 16042

Time Allowed: 03 Hours

Max. Marks: 60

NOTE:

- 1) Parts A and B are compulsory
- 2) Part-C has Two Questions Q8 and Q9. Both are compulsory, but with internal choice
- 3) Any missing data may be assumed appropriately

Part – A

[Marks: 02 each]

Q1.

- a) Convert $(725.25)_{10}$ to its decimal, binary
- b) Distinguish between weight code and non weighted code?
- c) Compare latch with flip flop.
- d) Outline the concept of duality in Boolean algebra.
- e) Distinguish between the synchronous and asynchronous counter?
- f) Is VHDL similar to C++.

Part – B

[Marks: 04 each]

Q2. Using NAND & NOR gate realize all the logic gate

Q3. Solve the Boolean expression using K-map in SOP method
 $Y = (WXYZ) = \sum M(0,2,4,6,8,10,12,13,15) \& d(1,5,7,9)$

Q4. Convert SR flip flop into JK flip flop.

Q5. Write the meaning of Excitation table, Edge triggering, Level triggering.

Q6. Design MOD 10 synchronous counter using JK Flip flop.

Q7. Discuss the working of Successive Approximation type converter.

MORNING

Part – C

01 MAR 2024

[Marks: 12 each]

- Q8. A staircase light is controlled by two switches , one is at the top of the stairs and the other is at the bottom of the stairs :
- (i) Make a truth table for this system.
 - (ii) Write the logic function in SOP form.
 - (iii) Realize the circuit using AOI logic.
 - (iv) Realize the circuit using minimum number of NAND and NOR gates.

OR

How do we classify sequential circuits? Also discuss the process to design a full adder using two half adders and an OR gate.

- Q9. Compare MUX and DMUX. Implement the following Boolean function using an 8:1 multiplexer considering D as the input and A,B,C as selection lines : $F(A, B, C, D) = AB' + BD + B'CD'$

OR

Elaborate the minimization of the given Boolean function using Quine-Mc-Cluskey method $F = m(0,1,2,5,7,8,9,10,13,15)$. Σ Realize the simplified function using logic gates.
